Highlights

Dynamic ICU Bed Allocation Under Resource Scarcity: A Discrete-Event Simulation Integrating Multi-Type Beds and Triage Policies

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- Dynamic simulation of four ICU bed pools calibrated with CNES and Epimed data.
- Models waiting-queue clinical decay and severity-dependent length of stay.
- Benchmarks cμ-type and austere indices against an offline LP/MILP upper bound.
- CMU-MGS saves 20.8 more expected lives than FCFS at $\rho = 1.6$ ($p_{\text{Holm}} < 0.001$).
- Quantifies adult–coronary overflow gains and robustness to clinical noise.

Dynamic ICU Bed Allocation Under Resource Scarcity: A Discrete-Event Simulation Integrating Multi-Type Beds and Triage Policies

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Abstract

During acute public health crises, epidemic surges, and mass-casualty incidents, intensive care unit (ICU) demand frequently exceeds physical bed availability. Conventional optimization models often formulate ICU triage as a static, single-batch portfolio selection problem with homogeneous beds, overlooking continuous stochastic arrivals, waiting-time clinical deterioration, severity-correlated length of stay (LOS), and specialty barriers across critical care units. This paper presents a continuous-time discrete-event simulation (DES) framework and a retrospective time-indexed mixed-integer linear programming (MILP) benchmark for dynamic multi-type ICU bed allocation. The model is calibrated using official bed capacities from the Brazilian National Register of Healthcare Establishments (DATASUS/CNES) for the University Hospital of the Federal University of Maranhão (HU-UFMA; 55 critical care beds across adult, coronary, pediatric, and neonatal units) combined with specialty-specific epidemiological parameters from the Brazilian ICU Registry (AMIB/Epimed) and the Brazilian Neonatal Research Network. We evaluate six online admission rules—First-Come, First-Served (FCFS), Most Serious First (MSF), Least Serious First (LSF), Marginal Gain in Survival (MGS), a benefit-per-bed-day index (CMU-MGS), and a deterioration-rate index (AUSTERE)—alongside an offline LP/MILP upper bound across 100 Common Random Numbers (CRN) replications over a 60-day steady-state horizon. Under severe congestion ($\rho = 1.60$), CMU-MGS and AUSTERE save 20.80 (95% CI $[20.09, 21.50]$, +4.82%, $p_{\text{Holm}} < 0.001$) and 20.13 (+4.67%) more expected lives than FCFS, respectively, while cutting mean waiting times of admitted patients from 18.3 to 9.6 hours and achieving an optimality gap of 9.93% relative to the clairvoyant upper bound. A two-parameter sensitivity analysis across severity–LOS coupling ($\beta \in [0, 2.4]$) and queue deterioration ($\alpha_{\text{det}} \in [0.05, 0.50]$ day^{-1}) reveals that normalizing marginal survival benefit by expected bed-days (CMU-MGS) saves up to +4.21 additional lives over greedy MGS ($p < 0.001$) when severe patients require substantially longer stays. Furthermore, hierarchical adult–coronary overflow pooling saves up to +3.90 expected lives at balanced capacity ($\rho = 1.00$), and index policies remain statistically superior to FCFS under simultaneous observation noise across all clinical and LOS inputs ($\sigma_{\text{noise}} \leq 0.20$).

Keywords: Healthcare Operations Research, Intensive Care Units, Discrete-Event Simulation, Triage Policies, Queueing with Abandonment, Mathematical Programming

1. Introduction

Intensive Care Units (ICUs) constitute a high-acuity, strictly capacity-constrained resource within tertiary healthcare systems. During epidemic waves, seasonal respiratory surges, or mass-casualty disasters, the arrival rate of critically ill patients requiring invasive organ support can rapidly exceed available bed capacity [1, 2]. Under acute scarcity, clinical leadership must rely on transparent triage protocols to determine which candidates are admitted to intensive care and which remain in general or transitional wards [3, 4, 5].

In day-to-day hospital routines, beds are predominantly assigned on a First-Come, First-Served (FCFS) basis. While simple to administer, chronological queueing ignores heterogeneity in baseline mortality risk, therapeutic benefit from intensive care, rate of physiological deterioration while waiting, and expected bed occupation duration [6, 7]. To support rational decision-making, recent studies have proposed portfolio optimization and multi-attribute utility models for ICU bed allocation [6, 8]. Although such models formalize prognostic scoring, they typically treat triage as a static, one-shot batch assignment in a single homogeneous bed pool.

In hospital operations, however, critical care allocation is governed by four interacting dynamic mechanisms:

(i) Asynchronous Arrivals and Bed Turnover: Patients arrive continuously over time, while bed vacancies arise stochastically as admitted patients recover, transfer to step-down wards, or die [9, 10, 11].

(ii) Waiting-Queue Clinical Deterioration and Abandon-

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ment: Candidates held in emergency departments or pre-ICU wards experience progressive physiological instability; prolonged boarding reduces the survival benefit of subsequent ICU admission and can culminate in pre-ICU mortality or clinical reneging [12, 7].

(iii) Severity–Duration Trade-off: More severely ill patients often require prolonged mechanical ventilation and vasoactive support, inducing a positive correlation between baseline clinical severity and ICU length of stay (LOS) [13, 14]. Consequently, admitting a patient with high absolute benefit may lock a bed for an extended period, blocking multiple shorter-stay candidates.

(iv) Multi-Type Specialization and Partial Pooling: Tertiary hospitals partition critical care into specialized units—neonatal, pediatric, general adult, and coronary care—due to equipment and clinical skill barriers [15, 16]. While pediatric and neonatal pools are strictly segregated, adult medical and coronary care units (UCO) can partially share compatible adult patients during surge conditions [17].

To capture these operational realities, this paper develops a continuous-time discrete-event simulation (DES) framework coupled with a retrospective mixed-integer linear programming (MILP) benchmark for multi-type ICU bed allocation. Our main contributions are:

- Dynamic Queueing Formulation with Deterioration and Severity-Coupled LOS: We model a multi-pool $M/G/c + G$ critical care system where post-admission survival probability decays exponentially with waiting time and individual expected LOS scales with baseline severity, creating an explicit operational tension between marginal survival gain and bed-day consumption.

- Empirical Multi-Pool Calibration: Physical bed capacities are parameterized from the Brazilian National Register of Healthcare Establishments (DATASUS/CNES) for the University Hospital of the Federal University of Maranhão (HU-UFMA: 55 beds across adult, coronary, pediatric, and neonatal units) [18], while specialty-specific LOS distributions, baseline survival ranges, and boarding decay rates are calibrated from large-scale Brazilian intensive care cohorts (AMIB/Epimed, ORCHESTRA, and the Brazilian Neonatal Research Network) [14, 13, 16, 12].

- Comprehensive Policy Benchmarking and Offline Optimality Gap: Using a 100-replication Common Random Numbers (CRN) design with steady-state warm-up and complete post-horizon cohort drainage, we compare six online rules—FCFS, Most Serious First (MSF), Least Serious First (LSF), Marginal Gain in Survival (MGS), a $c\mu$-type survival-rate index (CMU-MGS) [19], and an austere deterioration-rate index (AUSTERE) [7]—against a clairvoyant time-indexed LP/MILP upper bound.

- Parametric Regime Mapping, Pooling Ablation, and Imperfect Information Analysis: Through systematic sensitivity experiments across severity–LOS coupling (β), queue deterioration (α_{det}), adult–coronary overflow rules, and multi-parameter clinical estimation noise (σ_{noise}) [20], we characterize the exact operational conditions under which duration-aware index rules outperform greedy triage.

The remainder of this paper is organized as follows. Section 2 reviews the operations research literature on ICU management and dynamic triage. Section 3 formalizes the queueing model, triage indices, and offline MILP upper bound. Section 4 details the empirical parameterization from CNES and national ICU registries. Section 5 presents the simulation results, offline optimality gaps, sensitivity heatmaps, and ablation studies. Section 6 discusses clinical, ethical, and methodological limitations, and Section 7 concludes.

2. Related Work

The operations research and management science (OR/MS) literature on intensive care capacity management and patient prioritization spans empirical econometrics, Markov decision processes (MDPs), queueing theory with abandonment, and discrete-event simulation. Table 1 summarizes representative studies and contrasts their modeling assumptions with the present work.

A foundational stream of OR/MS research examines how ICU congestion shapes admission and discharge decisions. Chan et al. [9] formulated an optimization model for demand-driven early ICU discharge, Accounting for post-discharge readmission risks and deriving an index policy that prioritizes discharging patients with low marginal readmission cost per bed-day freed. Kim et al. [10] conducted an econometric and simulation study of 15 Kaiser Permanente hospitals, demonstrating that ICU occupancy at arrival significantly alters admission probability and that boarding delays measurably worsen patient outcomes. More recently, Valeva et al. [17] analyzed the interaction between ICUs and downstream step-down units under flexible staffing, deriving dual-based acuity policies to mitigate bed blocking, while Bilbao [11] and Murray et al. [23] combined survival analysis with arrival forecasting to predict short-term ICU census.

A second stream investigates casualty and critical care triage under acute resource scarcity, drawing heavily on the $c\mu$ scheduling rule [19] and queueing systems with impatient or deteriorating customers. Argon and Ziya [20] studied priority assignment when customer types are observed imperfectly, showing that classification noise degrades multi-class rules and favors robust index structures. Mills et al. [21] modeled mass-casualty prioritization

Table 1: Comparison of the proposed framework with representative ICU operations and triage models in the literature.

Study	Methodology	Bed Pools	Arrival Process	LOS Model	Queue Decay / Reneging	Triage / Control Policy	Offline Bound
Chan et al. [9]	MDP / Simulation	Single ICU	Stochastic	Geometric / Empirical	No (Readmission risk)	Discharge index policy	Dynamic prog.
Mills et al. [21]	Fluid / Queueing	Multi-resource	Static / Surge	Exp. / Resource-tied	Time-varying survival	Dynamic $c\mu$ / Myopic index	Fluid optimal
Kim et al. [10]	Econometric / Sim.	Single ICU	Empirical Poisson	Log-Normal / Empirical	Delayed admission cost	Threshold admission control	No
Sun et al. [7]	MDP / Queueing	Single server/pool	None / Clearing	Exponential (2 classes)	Discrete deterioration	Austere priority index	MDP optimal
Alban et al. [22]	Process DES	Single ICU	Poisson	Empirical / Weibull	Diversion / Overflow	Capacity planning threshold	No
Bilbao [11]	DES / Flight Sim.	Single ICU + Ward	Gompertz / Poisson	Empirical / Phase-type	Early discharge / Divert	Physician behavioral rules	No
Valeva et al. [17]	Two-stage LP / DES	ICU + Step-down	Markovian	Exponential	Bed blocking	Acuity-based outflow LP	Stationary dual
Frej et al. [8]	Static ILP / Utility	Single ICU	None (Batch)	Not modeled	No	Bayesian SOFA portfolio	Exact static ILP
Murray et al. [23]	Time-Series / Surv.	Single ICU	ARIMA	Cox / Weibull (MODS)	No	Census forecasting	No
This Work	DES + MILP Bound	4 Pools (Coupled)	Continuous Poisson	Severity-Log-Normal	Exp. Decay + Reneging	FCFS, MSF, LSF, MGS, CMU, AUSTERE	Time-indexed LP/MILP

across multiple bottleneck resources, showing that fluid-based dynamic indices accounting for both survival deterioration and resource processing rates outperform static color-tag triage (such as Simple Triage and Rapid Treatment). Closest to our clinical triage focus, Sun et al. [7] analyzed patient prioritization under austere conditions where queued patients transition between severity states and may abandon (die) before service. They proved that optimal prioritization balances the immediate marginal survival reward against expected service time and the hazard of clinical deterioration.

During the COVID-19 pandemic, several studies addressed hospital surge allocation. Alban et al. [22] built a discrete-event simulation (icu-covid-sim) to project regional overflow probabilities. White and Lo [5] and Aquino et al. [2] evaluated ethical multi-principle frameworks combining short-term prognosis with equity considerations. In the Brazilian context, Frej et al. [6, 8] proposed a binary portfolio optimization model using Expected Utility Theory and SOFA scores to select which candidates occupy open beds.

Despite these advances, Table 1 highlights an important methodological gap. Analytical triage models such as Sun et al. [7] and Mills et al. [21] rely on exponential service times and single-pool or clearing formulations to preserve mathematical tractability, whereas utility-based models like Frej et al. [8] omit continuous arrivals, bed turnover, and queue waiting decay entirely. Conversely, multi-unit hospital simulations typically apply simple FCFS or threshold rules without benchmarking $c\mu$-type indices or quantifying the optimality gap against a retrospective mathematical programming bound. Our work bridges this gap by embedding $c\mu$ and austere deterioration-rate indices within a continuous-time, multi-specialty hospital simulation calibrated with empirical Brazilian intensive care data, while evaluating their distance to an offline time-indexed LP/MILP upper bound.

3. Dynamic Simulation and Optimization Modeling

3.1. System Architecture and Queueing Dynamics

We consider a tertiary referral hospital operating over a continuous time horizon $[0, T_{\text{warm}} + T)$, where $T_{\text{warm}} > 0$ is an initial warm-up period used to reach stochastic steady state and $[T_{\text{warm}}, T_{\text{warm}} + T)$ is the active evaluation window of length T days. The critical care department comprises four specialized bed pools $J = \{\text{adult}, \text{coronary}, \text{pediatric}, \text{neonatal}\}$, with physical bed capacities $C_j \in \mathbb{Z}_+$ $(j \in J)$ and total capacity $C_{\text{total}} = \sum_{j \in J} C_j$.

Patients requiring intensive care in specialty $j \in J$ arrive according to an independent homogeneous Poisson process with rate $\lambda_j > 0$ (patients per day). Each arriving patient i is characterized by the attribute vector:

$$\boldsymbol{\xi}_i = \left(j(i), t^i_{\text{arr}}, P^i_{\text{out}}, P^i_{\text{in},0}, \alpha^{j(i)}_{\text{det}}, \mathbb{E}[\text{LOS}_i], \text{LOS}_i, W^i_{\max}, u_i\right) \quad (1)$$

where:

- $j(i) \in J$ is the patient's primary clinical specialty;
- $t^i_{\text{arr}} \geq 0$ is the arrival epoch;
- $P^i_{\text{out}} \in (0, 1)$ is the baseline probability of hospital survival if the patient is never admitted to an ICU bed (managed in general or emergency holding wards);
- $P^i_{\text{in},0} \in [P^i_{\text{out}}, 1)$ is the initial probability of survival if admitted immediately upon arrival $(w_i = 0)$, parameterized as $P^i_{\text{in},0} = P^i_{\text{out}} + (1 - P^i_{\text{out}})g_i$, where $g_i \in (0, 1)$ is the fractional therapeutic efficacy of ICU care;
- $\alpha^{j(i)}_{\text{det}} > 0$ is the specialty-specific clinical deterioration rate (day^{-1}) during pre-ICU queue boarding;
- $\mathbb{E}[\text{LOS}_i] > 0$ and $\text{LOS}_i > 0$ are the expected and realized ICU lengths of stay (in days) once admitted;
- $W^i_{\max} > 0$ is the finite queue waiting tolerance (deadline in days) before clinical reneging (pre-ICU death, emergency transfer, or loss of therapeutic window);
- $u_i \sim \text{Uniform}(0, 1)$ is an independent uniform random variate used to realize binary survival outcomes under Common Random Numbers.

3.1.1. Waiting-Time Clinical Deterioration

Clinical studies consistently show that each hour of delayed ICU transfer increases mortality risk due to progression of shock and organ dysfunction [12, 10]. If patient i waits $w = t - t^i_{\text{arr}} \in [0, W^i_{\max}]$ days in the pre-ICU queue before admission at time t, the post-admission survival probability decays exponentially toward the baseline floor survival P^i_{out}:

$$P^i_{\text{in}}(w) = P^i_{\text{out}} + \left(P^i_{\text{in},0} - P^i_{\text{out}}\right) \exp\left(-\alpha^{j(i)}_{\text{det}} w\right) \quad (2)$$

Consequently, the instantaneous marginal survival gain of admitting patient i after waiting time $w = t - t^i_{\text{arr}}$ is:

$$\Delta P_i(w) = P^i_{\text{in}}(w) - P^i_{\text{out}} = \left(P^i_{\text{in},0} - P^i_{\text{out}}\right) \exp\left(-\alpha^{j(i)}_{\text{det}} w\right) \quad (3)$$

3.1.2. Severity-Correlated Length of Stay

To reflect the clinical reality that patients with higher acute physiological severity require longer organ support [13, 14], the individual expected stay $\mathbb{E}[\mathrm{LOS}_i]$ is coupled to the patient's baseline mortality risk $1 - P^i_{\mathrm{out}}$:

$$\mathbb{E}[\mathrm{LOS}_i] = \mu_{\mathrm{los},j(i)} \cdot \max\Big\{0.35,\ 1 + \beta\Big((1 - P^i_{\mathrm{out}}) - \mathbb{E}[1 - P^{j(i)}_{\mathrm{out}}]\Big)\Big\} \tag{4}$$

where $\mu_{\mathrm{los},j}$ is the empirical mean LOS of specialty j, $\mathbb{E}[1 - P^j_{\mathrm{out}}]$ is the mean baseline severity in specialty j, and $\beta \geq 0$ governs the strength of the severity–duration coupling. Centering the multiplier around $\mathbb{E}[1 - P^j_{\mathrm{out}}]$ ensures that the unconditional mean stay across arriving patients in pool j equals $\mathbb{E}[\mathrm{LOS}_i] = \mu_{\mathrm{los},j}$, preserving exact traffic intensity scaling across all values of β.

Given $\mathbb{E}[\mathrm{LOS}_i]$ and the specialty's coefficient of variation $CV_{\mathrm{los},j}$, the realized stay LOS_i follows a Log-Normal distribution $\mathrm{LogNormal}(\mu_i, \sigma_j^2)$ with exact log-scale parameters:

$$\sigma_j = \sqrt{\ln\Big(1 + CV^2_{\mathrm{los},j}\Big)}, \qquad \mu_i = \ln(\mathbb{E}[\mathrm{LOS}_i]) - \frac{1}{2}\sigma_j^2 \tag{5}$$

3.1.3. Hierarchical Multi-Pool Compatibility and Overflow

Pediatric (j = pediatric) and neonatal (j = neonatal) units require dedicated ventilators, incubators, and subspecialty nursing, operating as strictly segregated pools. Conversely, general adult ICU beds (j = adult) and adult coronary care beds (j = coronary) share adult life-support infrastructure and can operate under a hierarchical partial-pooling (overflow) protocol:

(1) Primary Home-Pool Priority: When a bed in pool $j \in \{\text{adult}, \text{coronary}\}$ becomes vacant at time t, if the native queue $Q_j(t) \neq \emptyset$, the bed is assigned exclusively to the highest-priority candidate in $Q_j(t)$ according to active triage policy π.

(2) Secondary Cross-Pool Overflow: If the native queue is empty ($Q_j(t) = \emptyset$) and the coupled adult pool $j' \in \{\text{adult}, \text{coronary}\} \setminus \{j\}$ has waiting candidates ($Q_{j'}(t) \neq \emptyset$), the vacant bed in j admits the highest-priority patient from $Q_{j'}(t)$ under policy π. Similarly, an arriving adult or coronary patient finding their native unit full is immediately admitted to the partner unit if an idle bed is available there.

Because cross-pool admission occurs only when the home queue is empty, native and overflow candidates never compete simultaneously for the same bed vacancy; within any non-empty queue, candidates are ordered strictly by policy score $s^\pi_i(t)$.

3.2. Online Triage Policies

Whenever a bed vacancy occurs at epoch t in an eligible queue $Q(t) \neq \emptyset$, triage policy π admits patient i^*:

$$i^* = \arg\max_{i \in Q(t)} \left\{ s^\pi_i(t) \right\}, \quad \text{with ties broken by earliest } t^i_{\mathrm{arr}} \tag{6}$$

We evaluate six online prioritization policies:

1. First-Come, First-Served (FCFS): Prioritizes candidates in chronological order of arrival:

$$s^{\mathrm{FCFS}}_i(t) = -t^i_{\mathrm{arr}} \tag{7}$$

2. Most Serious First (MSF): Prioritizes the sickest candidates with the highest baseline mortality risk without ICU care:

$$s^{\mathrm{MSF}}_i(t) = 1 - P^i_{\mathrm{out}} \tag{8}$$

MSF prioritizes acute severity regardless of whether the patient has high therapeutic responsiveness or prolonged expected stay.

3. Least Serious First (LSF): Prioritizes candidates with the highest absolute post-ICU survival probability at decision time t:

$$s^{\mathrm{LSF}}_i(t) = P^i_{\mathrm{in}}(t - t^i_{\mathrm{arr}}) \tag{9}$$

4. Marginal Gain in Survival (MGS): Prioritizes candidates with the largest instantaneous incremental survival benefit from ICU admission:

$$s^{\mathrm{MGS}}_i(t) = \Delta P_i(t - t^i_{\mathrm{arr}}) = P^i_{\mathrm{in}}(t - t^i_{\mathrm{arr}}) - P^i_{\mathrm{out}} \tag{10}$$

5. Benefit-per-Bed-Day Index (CMU-MGS): Motivated by the classical $c\mu$ scheduling rule [19, 21], CMU-MGS normalizes the instantaneous marginal survival gain by the patient's expected bed-days $\mathbb{E}[\mathrm{LOS}_i]$:

$$s^{\text{CMU-MGS}}_i(t) = \frac{\Delta P_i(t - t^i_{\mathrm{arr}})}{\max\left(0.2,\ \mathbb{E}[\mathrm{LOS}_i]\right)} \tag{11}$$

CMU-MGS maximizes the rate of expected lives saved per unit of bottleneck bed capacity consumed.

6. Austere Deterioration-Rate Index (AUSTERE): Adapted from the austere-conditions triage theory of Sun et al. [7], this policy prioritizes patients according to the expected marginal survival loss incurred if admission is delayed, normalized by expected service duration $\mathbb{E}[\mathrm{LOS}_i]$. A waiting patient experiences both continuous exponential decay at rate $\alpha^{j(i)}_{\mathrm{det}}$ and an increasing hazard of hitting the reneging deadline $\tau_i(t) = \max\Big(0.25,\ t^i_{\mathrm{arr}} + W^i_{\max} - t\Big)$:

$$s^{\mathrm{AUST}}_i(t) = \frac{\Delta P_i(t - t^i_{\mathrm{arr}})}{\max(0.2, \mathbb{E}[\mathrm{LOS}_i])} \left(1 + \alpha^{j(i)}_{\mathrm{det}} + \frac{0.5}{\tau_i(t) + 0.5}\right) \tag{12}$$

Unlike unbounded reciprocal deadline rules, Eq. (12) bounds the urgency multiplier smoothly, balancing bed turnover efficiency against imminent clinical deterioration.

3.2.1. Triage Under Imperfect Clinical Information

In clinical practice, prognostic probabilities and expected stay durations are estimated from physiological scores (e.g., SAPS 3, SOFA, PIM 3, SNAPPE-II) subject to estimation error [20, 8]. To evaluate informational robustness fairly across all clinical policies, when observation noise $\sigma_{\text{noise}} > 0$ is present, the triage controller observes noisy estimates $(\widetilde{P}^i_{\text{out}}, \widetilde{P}^i_{\text{in},0}, \widetilde{\mathbb{E}}[\text{LOS}_i])$ for every patient:

$$\begin{aligned} \widetilde{P}^i_{\text{out}} &= \text{clip}_{[0.01,0.95]}\Big(P^i_{\text{out}} + \varepsilon_{i,1}\Big), \\ \widetilde{P}^i_{\text{in},0} &= \text{clip}_{[\widetilde{P}^i_{\text{out}},0.99]}\Big(P^i_{\text{in},0} + \varepsilon_{i,2}\Big), \\ \widetilde{\mathbb{E}}[\text{LOS}_i] &= \mathbb{E}[\text{LOS}_i] \cdot \max(0.4,\ 1 + \varepsilon_{i,3}) \end{aligned} \tag{13}$$

where $\varepsilon_{i,1}, \varepsilon_{i,2}, \varepsilon_{i,3} \sim \mathcal{N}(0, \sigma^2_{\text{noise}})$ are independent Gaussian perturbations. All non-FCFS policies (MSF, LSF, MGS, CMU-MGS, and AUSTERE) compute their priority scores using these noisy estimates, whereas actual post-triage survival outcomes are evaluated using the true underlying parameters $(P^i_{\text{out}}, P^i_{\text{in},0})$.

3.3. Evaluation Cohort, Post-Horizon Drainage, and Metrics

To eliminate initialization bias and boundary truncation distortions, each simulation replication generates arrivals over $[0, T_{\text{warm}}+T)$ and continues executing discharge, admission, and timeout events beyond $T_{\text{warm}} + T$ until every patient who arrived during the evaluation window $[T_{\text{warm}}, T_{\text{warm}} + T)$ has either been admitted to a bed or reneged from the queue. Let $\mathcal{I}_{\text{eval}} = \{i : T_{\text{warm}} \le t^i_{\text{arr}} < T_{\text{warm}} + T\}$ denote the evaluated patient cohort of size $N_{\text{eval}} = |\mathcal{I}_{\text{eval}}|$, partitioned into the admitted subset $\mathcal{A}^\pi \subseteq \mathcal{I}_{\text{eval}}$ and its complement (the reneged/unadmitted subset) $\mathcal{R}^\pi = \mathcal{I}_{\text{eval}} \setminus \mathcal{A}^\pi$. Because simulation continues until all queues drain, $\mathcal{A}^\pi \cup \mathcal{R}^\pi = \mathcal{I}_{\text{eval}}$ holds exactly.

System performance over $\mathcal{I}_{\text{eval}}$ is quantified by:

- Expected Survivors:

$$S^\pi_{\text{exp}} = \sum_{i \in \mathcal{A}^\pi} P^i_{\text{in}}\Big(t^i_{\text{adm}} - t^i_{\text{arr}}\Big) + \sum_{i \in \mathcal{R}^\pi} P^i_{\text{out}} \tag{14}$$

- Realized Survivors:

$$S^\pi_{\text{real}} = \sum_{i \in \mathcal{A}^\pi} \mathbb{I}\Big(u_i < P^i_{\text{in}}(t^i_{\text{adm}} - t^i_{\text{arr}})\Big) + \sum_{i \in \mathcal{R}^\pi} \mathbb{I}\Big(u_i < P^i_{\text{out}}\Big) \tag{15}$$

- Disaggregated Waiting Times and Reneging Rate: We report the reneging rate $r^\pi_{\text{ren}} = |\mathcal{R}^\pi|/N_{\text{eval}}$, the mean waiting time of admitted patients $\bar{W}^\pi_{\text{adm}} = \frac{24}{|\mathcal{A}^\pi|} \sum_{i \in \mathcal{A}^\pi} (t^i_{\text{adm}} - t^i_{\text{arr}})$ (in hours), and the mean waiting time prior to abandonment for reneged patients $\bar{W}^\pi_{\text{ren}} = \frac{24}{|\mathcal{R}^\pi|} \sum_{i \in \mathcal{R}^\pi} W^i_{\max}$.

- Steady-State Time-Weighted Occupancy: Measured strictly over the steady-state observation window $[T_{\text{warm}}, T_{\text{warm}} + T)$:

$$\bar{O}^\pi_j = \frac{1}{T \cdot C_j} \int_{T_{\text{warm}}}^{T_{\text{warm}}+T} B^\pi_j(t)\, dt \tag{16}$$

where $B^\pi_j(t) \in \{0, 1, \ldots, C_j\}$ is the number of occupied beds in pool j at time t.

3.4. Hindsight Time-Indexed MILP Upper Bound

To measure how close online heuristics come to the theoretical maximum survival achievable under complete future knowledge, we formulate a clairvoyant (offline) time-indexed mathematical program for each replication. Let time be discretized into slots $s \in \mathcal{S} = \{0, 1, \ldots, S_{\max}\}$ of length $\Delta t = 0.5$ days. For each pool group g with capacity C_g and each patient i arriving at slot $s^i_{\text{arr}} = \lfloor t^i_{\text{arr}}/\Delta t \rfloor$ with stay duration $L_i = \max(1, \lfloor \text{LOS}_i/\Delta t \rfloor)$ slots and maximum wait $K_i = \lceil W^i_{\max}/\Delta t \rceil$ slots, let binary decision variable $x_{i,k} \in \{0, 1\}$ indicate whether patient i is admitted after waiting $k \in \{0, 1, \ldots, K_i\}$ slots (at slot $s = s^i_{\text{arr}} + k$). The hindsight upper bound S_{UB} is obtained by solving:

$$\max_x \quad \sum_{i \in \mathcal{I}_{\text{eval}}} P^i_{\text{out}} + \sum_{i \in \mathcal{I}_{\text{eval}}} \sum_{k=0}^{K_i} \Delta P_i\Big(\min(W^i_{\max}, k\Delta t)\Big)\, x_{i,k} \tag{17}$$

$$\text{s.t.} \quad \sum_{k=0}^{K_i} x_{i,k} \le 1, \qquad \forall i \tag{18}$$

$$\sum_{i \in \mathcal{I}_g} \sum_{\substack{k=0 \\ s - L_i < s^i_{\text{arr}} + k \le s}}^{K_i} x_{i,k} \le C_g, \qquad \forall g,\ \forall s \in \mathcal{S} \tag{19}$$

$$x_{i,k} \in [0, 1], \qquad \forall i, k \tag{20}$$

Constraint (18) ensures each patient is admitted at most once within their clinical tolerance window, while Constraint (19) enforces bed capacity C_g across all active time slots s (including boundary occupation from warm-up arrivals). Solving the linear programming relaxation (17)–(20) via the HiGHS interior-point/simplex solver yields a valid, strict upper bound $S_{\text{UB}} \ge S^\pi_{\text{exp}}$ on any online or offline non-preemptive policy.

4. Experimental Design and Empirical Calibration

4.1. Hospital Bed Capacities and Epidemiological Calibration

To ground the simulation in real-world public healthcare operations without relying on arbitrary synthetic parameters, we combine facility-level bed data from the University Hospital of the Federal University of Maranhão (HU-UFMA / EBSERH) with specialty-specific clinical parameters from multicenter Brazilian intensive care registries (Table 2).

According to the Brazilian National Register of Healthcare Establishments (DATASUS/CNES, Establishment ID 2111302726653, competency 12/2024) [18], HU-UFMA operates $C_{\text{total}} = 55$ permanent critical care beds dedicated exclusively to the Unified Health System (SUS): 15 Adult Type III ICU beds, 10 Type III Coronary Care (UCO) beds, 10 Pediatric Type III ICU beds, and 20 Neonatal Type II ICU beds.

Table 2: Empirical multi-pool calibration from DATASUS/CNES (HU-UFMA, competency 12/2024) and national Brazilian intensive care registries.

Bed Pool (j)	CNES Code / Specialty	Beds (C_j)	Mean LOS ($\mu_{\text{los},j}$)	LOS CV_j	Baseline Survival (P^i_{out})	ICU Efficacy (g_i)	Queue Decay (α^j_{det}, $W^i_{\max}$)
Adult ICU	76 – UTI Adulto Tipo III	15	6.5 days	0.65	$U(0.10, 0.55)$	$U(0.25, 0.75)$	0.18 d^{-1}, $U(0.5, 2.5)$ d
Coronary (UCO)	86 – UCO Tipo III	10	4.8 days	0.55	$U(0.20, 0.65)$	$U(0.30, 0.80)$	0.20 d^{-1}, $U(0.5, 2.0)$ d
Pediatric ICU	79 – UTI Pediátrica Tipo III	10	6.0 days	0.65	$U(0.25, 0.70)$	$U(0.35, 0.85)$	0.15 d^{-1}, $U(0.5, 2.5)$ d
Neonatal ICU	81 – UTI Neonatal Tipo II	20	11.5 days	0.75	$U(0.15, 0.65)$	$U(0.30, 0.85)$	0.20 d^{-1}, $U(0.5, 2.0)$ d
Total / System	Estab. CNES 2111302726653 (100% SUS)	**55**	Empirical sources: AMIB/Epimed [14, 13], RBPN [16], Cardoso et al. [12]				

Clinical distributions are differentiated across the four pools according to published Brazilian epidemiological cohorts:

- Adult and Coronary Units: In the national AMIB/Epimed and ORCHESTRA registries covering over 129,000 Brazilian ICU admissions [13, 14], public-sector adult ICUs exhibit mean stays of 6.0–7.5 days ($\mu_{\text{los, adult}} = 6.5$ days, $CV = 0.65$) and hospital mortality rates near 27% for admitted patients, whereas coronary units exhibit shorter mean stays ($\mu_{\text{los, coronary}} = 4.8$ days, $CV = 0.55$). Pre-ICU boarding decay ($\alpha_{\text{det}} \in [0.18, 0.20]$ day^{-1} and $W_{\max} \in [0.5, 2.5]$ days) is calibrated from the Brazilian cohort study of Cardoso et al. [12], which documented a continuous rise in mortality risk across 24–72 hours of ICU boarding delay.

- Pediatric and Neonatal Units: Based on AMIB pediatric data and the Brazilian Neonatal Research Network (RBPN) [16], pediatric ICU stays average $\mu_{\text{los, pediatric}} = 6.0$ days with higher post-admission survival, whereas neonatal ICU patients (including preterm and very-low-birth-weight infants) require substantially longer hospital stays ($\mu_{\text{los, neonatal}} = 11.5$ days, $CV = 0.75$). In our baseline configuration, the severity–LOS coupling parameter in Eq. (4) is set to $\beta = 1.2$, inducing a 2.1× variation in expected stay between the mildest and most severe arrivals within each pool.

4.2. Effective Traffic Intensity and Simulation Protocol

Because Eq. (4) centers the severity multiplier around $\mathbb{E}[1 - P^j_{\text{out}}]$, the unconditional mean LOS in each pool j is exactly $\mathbb{E}[\text{LOS}_i] = \mu_{\text{los},j}$. By Little's Law, to achieve a uniform effective traffic intensity ρ across all four specialized units, the Poisson arrival rate for pool $j \in J$ is set to:

$$\lambda_j(\rho) = \frac{\rho \cdot C_j}{\mu_{\text{los},j}}, \qquad \forall j \in J \tag{21}$$

Thus, the nominal parameter ρ equals the true effective traffic intensity $\rho_{\text{eff}} = \lambda_j \mathbb{E}[\text{LOS}_j]/C_j = \rho$ in every pool. Over the $T = 60$-day evaluation window, the total expected arrival volume across all 55 beds is $60 \sum_{j \in J} \lambda_j(\rho) = 468.2\,\rho$ patients (yielding on average 329.9, 467.6, 605.2, and 744.7 evaluated arrivals at $\rho \in \{0.70, 1.00, 1.30, 1.60\}$, respectively).

Each replication simulates $T_{\text{warm}} = 15$ warm-up days followed by $T = 60$ evaluation days, continuing execution after day 75 until all patients arriving in $[15, 75)$ complete service or renege. To guarantee zero cross-contamination between random inputs under Common Random Numbers, SHA-256 hash-derived seeds (stable_seed) initialize five independent Mersenne Twister streams per replication: (1) inter-arrival times, (2) clinical survival attributes, (3) Log-Normal LOS realizations, (4) uniform survival outcomes u_i, and (5) clinical observation noise ε_i. For statistical inference across the 20 policy-vs-FCFS comparisons (4 regimes×5 policies), we report paired means, standard errors, 95% confidence intervals, relative percentage gains, paired t-statistics, and Holm–Bonferroni step-down adjusted p-values (p_{Holm}).

5. Results and Discussion

5.1. Overall Performance and Hindsight Optimality Gap

Table 3 reports the steady-state operational and survival metrics across all four congestion regimes ($\rho \in \{0.70, 1.00, 1.30, 1.60\}$), alongside the clairvoyant LP/MILP upper bound (S_{UB}). Figure 1 illustrates the trajectory of expected survivors across traffic intensities relative to the hindsight upper bound.

Several key operational patterns emerge from Table 3:

(1) Consistency with Little's Law: Because arrival rates $\lambda_j(\rho)$ are calibrated to the exact pool-specific mean stay $\mu_{\text{los},j}$ and occupancy is integrated strictly over $[15, 75)$ days, time-weighted bed occupancy aligns closely with queueing theory. At $\rho = 0.70$, only 1.3%–1.4% of arrivals renege, yielding a steady-state occupancy of 68.8%–68.9% (matching the theoretical $M/G/c+G$ load $\rho(1 - r_{\text{ren}}) \approx 0.70 \times 0.986 = 69.0\%$). At $\rho = 1.00$, stochastic fluctuations under finite patience induce a 9.8%–10.8% reneging rate and 88.7%–89.4% bed utilization. Under severe surge ($\rho = 1.60$), the hospital operates near full saturation (98.6%–99.0% occupancy).

(2) Bed Turnover vs. Clinical Severity across Policies: The "Admitted LOS" column directly reflects the severity–duration coupling of Eq. (4). At $\rho = 1.60$, MSF prioritizes the sickest candidates ($1 - P^i_{\text{out}}$), resulting in the longest mean admitted stay (7.43 days) and the lowest admission volume (442.4 patients,

Table 3: Dynamic simulation performance across congestion regimes (60-day steady-state window after 15-day warm-up, 100 CRN replications; mean ± standard error).

Regime (ρ)	Policy	Total Arrivals	Admitted	Reneging Rate	Admitted Wait (h)	Reneged Wait (h)	Admitted LOS (d)	Bed Occupancy	Realized Survivors	Expected Survivors
0.70	FCFS	329.9 ± 1.7	325.5 ± 1.6	1.3% ± 0.1%	0.74 ± 0.06	21.8 ± 1.1	7.04 ± 0.05	68.9% ± 0.4%	237.3 ± 1.3	238.54 ± 1.25
	MSF	329.9 ± 1.7	325.1 ± 1.6	1.4% ± 0.1%	0.60 ± 0.05	24.2 ± 1.1	7.05 ± 0.05	68.9% ± 0.4%	237.4 ± 1.3	238.64 ± 1.25
	LSF	329.9 ± 1.7	325.2 ± 1.6	1.4% ± 0.1%	0.58 ± 0.05	24.1 ± 1.2	7.04 ± 0.05	68.9% ± 0.4%	237.3 ± 1.3	238.59 ± 1.25
	MGS	329.9 ± 1.7	325.1 ± 1.6	1.4% ± 0.1%	0.59 ± 0.05	24.7 ± 1.2	7.04 ± 0.05	68.9% ± 0.4%	237.3 ± 1.3	238.71 ± 1.25
	CMU-MGS	329.9 ± 1.7	325.2 ± 1.6	1.4% ± 0.1%	0.58 ± 0.05	23.8 ± 1.1	7.04 ± 0.05	68.8% ± 0.4%	237.3 ± 1.3	238.71 ± 1.25
	AUSTERE	329.9 ± 1.7	325.4 ± 1.6	1.4% ± 0.1%	0.61 ± 0.05	24.7 ± 1.2	7.04 ± 0.05	68.9% ± 0.4%	237.4 ± 1.3	**238.75 ± 1.25**
	Hindsight UB	329.9 ± 1.7	329.8 ± 1.7	—	—	—	—	—	—	*239.96 ± 1.25*
1.00	FCFS	467.6 ± 2.2	421.3 ± 1.8	9.8% ± 0.3%	5.65 ± 0.19	23.5 ± 0.3	7.06 ± 0.04	89.4% ± 0.3%	321.4 ± 1.5	320.23 ± 1.43
	MSF	467.6 ± 2.2	416.7 ± 1.8	10.8% ± 0.3%	4.20 ± 0.14	27.5 ± 0.3	7.12 ± 0.04	89.1% ± 0.3%	323.4 ± 1.5	321.90 ± 1.45
	LSF	467.6 ± 2.2	420.4 ± 1.8	10.0% ± 0.3%	3.82 ± 0.13	28.0 ± 0.4	7.03 ± 0.04	88.7% ± 0.3%	323.3 ± 1.5	322.24 ± 1.45
	MGS	467.6 ± 2.2	418.0 ± 1.8	10.5% ± 0.3%	3.87 ± 0.13	28.1 ± 0.4	7.09 ± 0.04	89.0% ± 0.3%	324.5 ± 1.5	323.37 ± 1.45
	AUSTERE	467.6 ± 2.2	419.4 ± 1.8	10.2% ± 0.3%	4.25 ± 0.14	28.9 ± 0.3	7.07 ± 0.04	89.0% ± 0.3%	324.3 ± 1.5	323.37 ± 1.45
	CMU-MGS	467.6 ± 2.2	419.0 ± 1.8	10.3% ± 0.3%	3.84 ± 0.13	28.0 ± 0.3	7.06 ± 0.04	88.8% ± 0.3%	**324.7 ± 1.5**	**323.39 ± 1.45**
	Hindsight UB	467.6 ± 2.2	455.3 ± 1.9	—	—	—	—	—	—	*335.16 ± 1.50*
1.30	FCFS	605.2 ± 2.2	453.4 ± 1.6	25.0% ± 0.3%	12.59 ± 0.17	24.8 ± 0.2	7.07 ± 0.04	96.8% ± 0.1%	377.2 ± 1.5	378.23 ± 1.29
	MSF	605.2 ± 2.2	440.9 ± 1.6	27.1% ± 0.3%	8.50 ± 0.12	29.1 ± 0.2	7.30 ± 0.04	96.7% ± 0.1%	383.2 ± 1.5	383.91 ± 1.32
	LSF	605.2 ± 2.2	457.4 ± 1.6	24.3% ± 0.3%	7.39 ± 0.10	29.3 ± 0.2	6.97 ± 0.04	96.1% ± 0.2%	384.8 ± 1.6	386.17 ± 1.36
	MGS	605.2 ± 2.2	445.4 ± 1.6	26.3% ± 0.3%	7.74 ± 0.11	29.4 ± 0.2	7.21 ± 0.04	96.5% ± 0.1%	387.8 ± 1.5	388.68 ± 1.34
	CMU-MGS	605.2 ± 2.2	449.4 ± 1.6	25.7% ± 0.3%	7.46 ± 0.10	29.5 ± 0.2	7.13 ± 0.04	96.4% ± 0.1%	388.2 ± 1.5	389.17 ± 1.34
	AUSTERE	605.2 ± 2.2	450.3 ± 1.6	25.5% ± 0.3%	8.37 ± 0.11	30.5 ± 0.2	7.12 ± 0.04	96.5% ± 0.1%	**388.7 ± 1.6**	**389.22 ± 1.35**
	Hindsight UB	605.2 ± 2.2	517.8 ± 1.6	—	—	—	—	—	—	*420.66 ± 1.35*
1.60	FCFS	744.7 ± 2.5	462.0 ± 1.5	37.9% ± 0.3%	18.27 ± 0.16	25.5 ± 0.1	7.09 ± 0.04	99.0% ± 0.1%	430.2 ± 1.4	431.14 ± 1.13
	MSF	744.7 ± 2.5	442.4 ± 1.5	40.5% ± 0.3%	11.09 ± 0.11	29.9 ± 0.1	7.43 ± 0.04	98.9% ± 0.1%	442.0 ± 1.4	442.48 ± 1.18
	LSF	744.7 ± 2.5	473.4 ± 1.5	36.4% ± 0.3%	9.47 ± 0.09	30.1 ± 0.1	6.92 ± 0.04	98.6% ± 0.1%	446.4 ± 1.5	446.47 ± 1.23
	MGS	744.7 ± 2.5	451.8 ± 1.4	39.3% ± 0.3%	9.89 ± 0.10	30.3 ± 0.1	7.28 ± 0.04	98.8% ± 0.1%	451.2 ± 1.5	451.12 ± 1.23
	AUSTERE	744.7 ± 2.5	458.6 ± 1.5	38.4% ± 0.3%	10.89 ± 0.10	31.5 ± 0.1	7.17 ± 0.04	98.9% ± 0.1%	**451.6 ± 1.5**	451.27 ± 1.25
	CMU-MGS	744.7 ± 2.5	459.1 ± 1.5	38.3% ± 0.3%	9.56 ± 0.10	30.4 ± 0.1	7.16 ± 0.04	98.7% ± 0.1%	**451.6 ± 1.5**	**451.93 ± 1.25**
	Hindsight UB	744.7 ± 2.5	556.4 ± 1.5	—	—	—	—	—	—	*501.81 ± 1.25*

40.5% reneging). Conversely, LSF favors milder patients with high P^i_{in}, achieving the shortest mean stay (6.92 days) and admitting the highest number of patients (473.4, 36.4% reneging). Despite admitting 14.3 more patients than CMU-MGS, LSF saves fewer lives (446.47 vs. 451.93) because many of its admitted patients already had high baseline survival P^i_{out} in general wards.

(3) Distance to the Hindsight Upper Bound: As shown in Figure 1, the offline LP/MILP upper bound S_{UB} achieves 239.96, 335.16, 420.66, and 501.81 expected survivors across $\rho \in \{0.7, 1.0, 1.3, 1.6\}$. The corresponding optimality gap $(S_{\text{UB}} - S^{\text{CMU-MGS}}_{\text{exp}})/S_{\text{UB}}$ of the online CMU-MGS rule is 0.52% at $\rho = 0.70$, 3.51% at $\rho = 1.00$, 7.47% at $\rho = 1.30$, and 9.93% at $\rho = 1.60$. Note that the hindsight relaxation knows exact future realized stays LOS_i and future arrival epochs, allowing it to selectively cherry-pick ultra-short-stay realizations (556.4 admissions vs. 459.1 for CMU-MGS at $\rho = 1.60$). Even without clairvoyance, online CMU-MGS captures 29.3% of the total theoretical survival gap between FCFS and the offline LP relaxation under extreme surge.

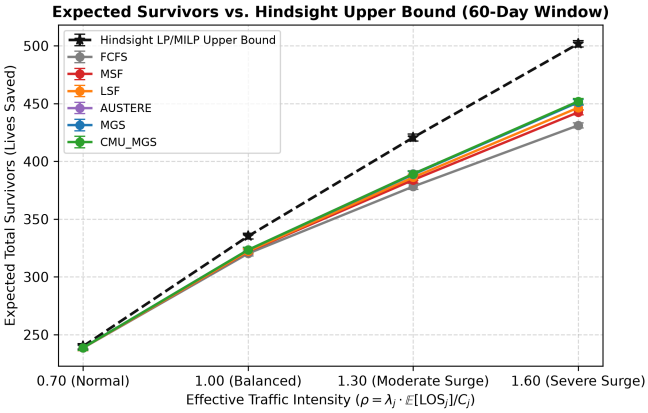


Figure 1: Expected total survivors across online triage policies compared against the hindsight time-indexed LP/MILP upper bound (S_{UB}) across effective traffic intensities $\rho \in \{0.7, 1.0, 1.3, 1.6\}$ (95% confidence intervals).

5.2. Paired Statistical Comparisons Over the FCFS Baseline

Table 4 and Figure 2 report the Common Random Numbers paired survival differences $\Delta S^{\pi}_r = S^{\pi}_{\text{exp},r} - S^{\text{FCFS}}_{\text{exp},r}$ across all 20 policy-regime pairs, adjusted for multiple testing via the Holm–Bonferroni procedure.

At low traffic intensity ($\rho = 0.70$), because variance reduction under CRN is substantial, small differences in how occasional bursts of arrivals are ordered produce statistically detectable t-statistics for MGS, CMU-MGS, and AUSTERE ($p_{\text{Holm}} < 0.001$), whereas LSF is statistically indistinguishable from FCFS ($p_{\text{Holm}} = 0.167$). In practical operational terms, however, the effect size at $\rho = 0.70$ is negligible across all rules ($\Delta S \leq 0.21$ lives over 60 days, or $< 0.09\%$ relative survival gain).

As traffic intensity reaches moderate ($\rho = 1.30$) and severe ($\rho = 1.60$) scarcity, clinically meaningful differences emerge. At $\rho = 1.60$, CMU-MGS saves $+20.80 \pm 0.36$ more expected lives than FCFS (+4.82%, $p_{\text{Holm}} < 0.001$), outperforming AUSTERE ($+20.13 \pm 0.38$), greedy MGS ($+19.98 \pm 0.34$), LSF ($+15.33 \pm 0.36$), and MSF ($+11.35 \pm 0.38$). In a direct paired test at $\rho = 1.60$ under the baseline parameter $\beta = 1.2$, CMU-MGS outperforms greedy MGS by

Table 4: Paired expected survival differences relative to the FCFS hospital baseline across 100 CRN replications (mean ± SE, 95% paired CI, relative gain, paired t-statistic, and Holm–Bonferroni adjusted p-value across 20 tests).

Regime (ρ)	Policy vs. FCFS	Paired Difference (ΔS)	95% Paired CI	Relative Effect	t-statistic	p_{Holm}
0.70	LSF vs. FCFS	+0.05 ± 0.03	[−0.02, +0.12]	+0.02%	1.39	0.167
	MSF vs. FCFS	+0.10 ± 0.03	[+0.03, +0.16]	+0.04%	2.92	0.007
	MGS vs. FCFS	+0.17 ± 0.04	[+0.09, +0.24]	+0.07%	4.14	< 0.001
	CMU-MGS vs. FCFS	+0.17 ± 0.04	[+0.09, +0.25]	+0.07%	4.16	< 0.001
	AUSTERE vs. FCFS	**+0.21 ± 0.04**	[+0.13, +0.29]	+0.09%	5.35	< 0.001
1.00	MSF vs. FCFS	+1.67 ± 0.14	[+1.40, +1.95]	+0.52%	11.79	< 0.001
	LSF vs. FCFS	+2.01 ± 0.15	[+1.71, +2.31]	+0.63%	13.14	< 0.001
	MGS vs. FCFS	+3.13 ± 0.17	[+2.81, +3.46]	+0.98%	18.90	< 0.001
	AUSTERE vs. FCFS	+3.14 ± 0.17	[+2.81, +3.47]	+0.98%	18.78	< 0.001
	CMU-MGS vs. FCFS	**+3.16 ± 0.17**	[+2.83, +3.49]	+0.99%	18.80	< 0.001
1.30	MSF vs. FCFS	+5.68 ± 0.26	[+5.16, +6.20]	+1.50%	21.46	< 0.001
	LSF vs. FCFS	+7.94 ± 0.28	[+7.39, +8.50]	+2.10%	28.17	< 0.001
	MGS vs. FCFS	+10.45 ± 0.30	[+9.87, +11.04]	+2.76%	35.11	< 0.001
	CMU-MGS vs. FCFS	+10.94 ± 0.27	[+10.40, +11.48]	+2.89%	39.81	< 0.001
	AUSTERE vs. FCFS	**+10.99 ± 0.33**	[+10.35, +11.63]	+2.91%	33.56	< 0.001
1.60	MSF vs. FCFS	+11.35 ± 0.38	[+10.59, +12.10]	+2.63%	29.49	< 0.001
	LSF vs. FCFS	+15.33 ± 0.36	[+14.63, +16.03]	+3.56%	43.01	< 0.001
	MGS vs. FCFS	+19.98 ± 0.34	[+19.31, +20.65]	+4.63%	58.37	< 0.001
	AUSTERE vs. FCFS	+20.13 ± 0.38	[+19.38, +20.89]	+4.67%	52.35	< 0.001
	CMU-MGS vs. FCFS	**+20.80 ± 0.36**	[+20.09, +21.50]	+4.82%	58.06	< 0.001

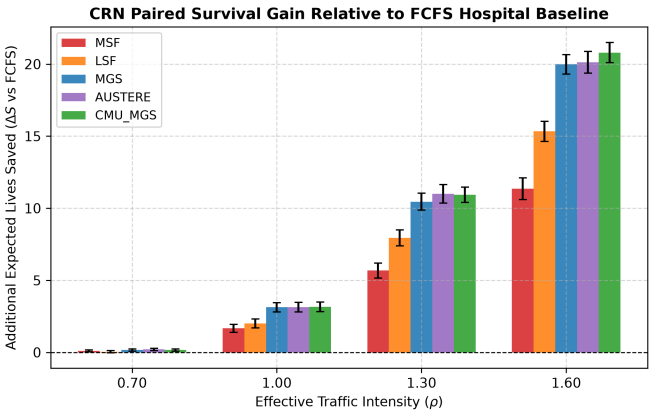


Figure 2: Paired incremental expected lives saved (ΔS) over the FCFS baseline across traffic intensities $\rho \in \{0.7, 1.0, 1.3, 1.6\}$ (95% paired confidence intervals).

+0.81 ± 0.25 expected lives ($t = 3.25, p = 0.0016$), confirming that normalizing marginal benefit by expected stay improves survival even under moderate severity–LOS correlation.

5.3. Disaggregated Waiting Times, Per-Pool Performance, and Overflow Ablation

Figure 3 displays the mean waiting time of admitted patients (Panel A) and the queue reneging rate (Panel B) across congestion levels. Under chronological FCFS at $\rho = 1.60$, every newly arriving candidate must wait behind the entire backlog of earlier arrivals, driving the mean waiting time of admitted patients to 18.27 hours and exposing all admitted patients to substantial exponential survival decay under Eq. (2). Under non-FCFS priority rules (CMU-MGS, MGS, LSF, AUSTERE, and MSF), high-priority candidates bypass the queue upon bed release, reducing the mean wait of admitted patients to 9.47–11.09 hours while lower-priority patients remain at the back of the queue until reaching their reneging deadline (30.1–31.5 hours). Notably, the principled AUSTERE index (Eq. 12) maintains low admitted waiting times (10.89 hours at $\rho = 1.60$), avoiding the pathological deadline-chasing behavior of unbounded reciprocal urgency weights.

Table 5 reports the disaggregated admissions and occupancy rates across the four specialized pools at $\rho = 1.00$ and $\rho = 1.60$, together with the paired ablation of the hierarchical adult–coronary overflow mechanism ($\Delta S_{\text{overflow}} = S_{\text{exp}}^{\text{overflow ON}} - S_{\text{exp}}^{\text{overflow OFF}}$).

The ablation study in Table 5 reveals a classic queueing pooling phenomenon. Cross-unit overflow between the 15-bed Adult ICU and 10-bed Coronary Unit is most impactful at balanced traffic ($\rho = 1.00$), where asynchronous random bursts temporarily saturate one unit while the partner unit has idle beds: enabling overflow admits +11.78 additional patients and saves +3.46 ± 0.14 expected lives under CMU-MGS ($t = 25.38, p < 0.001$) and +3.90 ± 0.18 lives under FCFS. Significant gains also occur at $\rho = 0.70$ (+1.40 lives, $p < 0.001$) and $\rho = 1.30$ (+1.56 lives, $p < 0.001$). Conversely, at extreme overload ($\rho = 1.60$), both the adult and coronary pools are continuously saturated (99.6%–99.8% occupancy) with persistent native queues, leaving almost no idle capacity for cross-unit sharing ($\Delta S = +0.39 \pm 0.20, p = 0.052$).

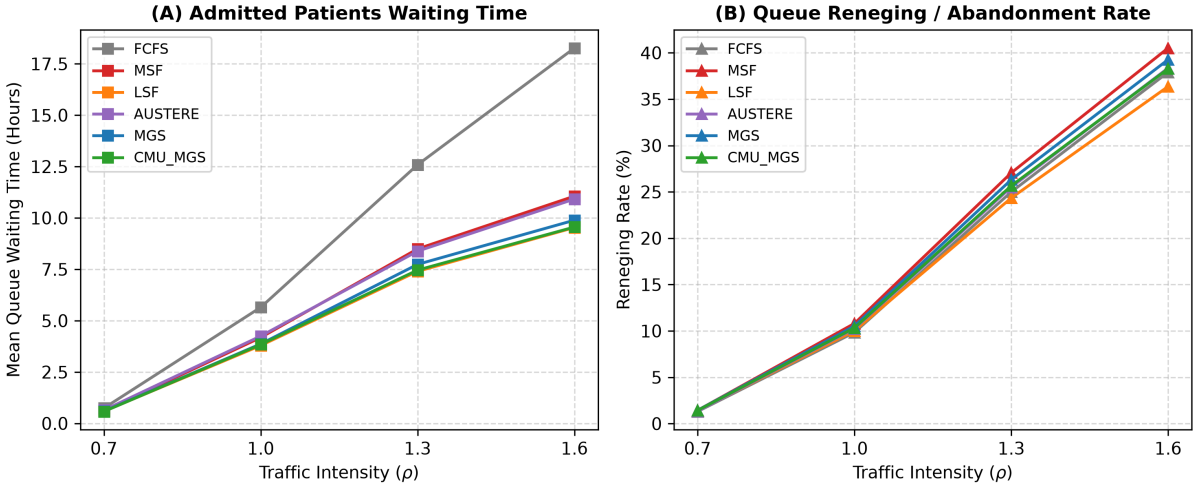


Figure 3: Queueing dynamics across traffic intensities ρ: (A) mean waiting time of admitted patients (hours); (B) percentage of arriving patients who renege before ICU admission.

Table 5: Disaggregated per-pool admissions and occupancy (mean over 100 replications) and paired ablation of adult–coronary overflow pooling.

		Disaggregated Admissions / Arrivals (Pool Occupancy %)				Adult–Coronary Overflow Ablation (ON minus OFF)		
Regime (ρ)	Policy	Adult (15 beds)	Coronary (10 beds)	Pediatric (10 beds)	Neonatal (20 beds)	Δ Admissions	Δ Expected Survivors	Paired t (p-value)
0.70	FCFS	96.9/97.9 (71.0%)	88.0/88.8 (71.3%)	69.2/70.1 (67.8%)	71.4/73.1 (66.8%)	+3.58 ± 0.22	+1.52 ± 0.10	t = 15.08 (p < 0.001)
	MGS	96.8/97.9 (70.9%)	87.9/88.8 (71.2%)	69.1/70.1 (67.7%)	71.3/73.1 (66.8%)	+4.01 ± 0.23	+1.42 ± 0.09	t = 15.87 (p < 0.001)
	CMU-MGS	96.8/97.9 (70.9%)	88.0/88.8 (71.1%)	69.1/70.1 (67.6%)	71.3/73.1 (66.7%)	+3.93 ± 0.23	+1.40 ± 0.09	t = 15.38 (p < 0.001)
1.00	FCFS	129.8/139.8 (92.4%)	113.0/124.3 (91.6%)	85.9/99.4 (86.3%)	92.5/104.1 (87.6%)	+11.16 ± 0.46	+3.90 ± 0.18	t = 21.34 (p < 0.001)
	MGS	129.2/139.8 (92.0%)	112.0/124.3 (91.0%)	84.8/99.4 (85.8%)	92.1/104.1 (87.2%)	+11.81 ± 0.45	+3.40 ± 0.14	t = 24.94 (p < 0.001)
	CMU-MGS	129.2/139.8 (91.9%)	112.3/124.3 (90.8%)	85.0/99.4 (85.6%)	92.4/104.1 (87.2%)	+11.78 ± 0.44	+3.46 ± 0.14	t = 25.38 (p < 0.001)
1.30	FCFS	139.6/181.2 (98.5%)	117.1/160.7 (98.3%)	94.9/128.8 (94.8%)	101.8/134.5 (95.6%)	+5.99 ± 0.46	+1.30 ± 0.19	t = 6.98 (p < 0.001)
	MGS	137.3/181.2 (98.3%)	114.3/160.7 (98.0%)	93.2/128.8 (94.3%)	100.6/134.5 (95.2%)	+7.17 ± 0.45	+1.49 ± 0.17	t = 8.54 (p < 0.001)
	CMU-MGS	138.9/181.2 (98.2%)	115.6/160.7 (97.9%)	93.9/128.8 (94.2%)	101.1/134.5 (95.1%)	+7.17 ± 0.46	+1.56 ± 0.19	t = 8.26 (p < 0.001)
1.60	FCFS	142.9/222.1 (99.8%)	118.3/198.1 (99.8%)	97.5/158.8 (98.0%)	103.3/165.7 (98.4%)	+0.44 ± 0.40	−0.31 ± 0.21	t = −1.46 (p = 0.145)
	MGS	140.9/222.1 (99.7%)	115.1/198.1 (99.7%)	95.3/158.8 (97.8%)	100.4/165.7 (98.2%)	+1.87 ± 0.40	+0.27 ± 0.18	t = 1.51 (p = 0.132)
	CMU-MGS	143.6/222.1 (99.7%)	117.5/198.1 (99.6%)	96.0/158.8 (97.7%)	102.1/165.7 (98.0%)	+2.72 ± 0.42	+0.39 ± 0.20	t = 1.95 (p = 0.052)

5.4. Sensitivity Analysis: Imperfect Information and Severity–LOS Regimes

To address how informational errors and clinical parameter variations affect policy rankings, Figure 4 presents two complementary sensitivity experiments at $\rho = 1.60$: (Panel A) simultaneous observation noise $\sigma_{\text{noise}} \in \{0.00, 0.05, 0.10, 0.20\}$ applied to P^i_{out}, $P^i_{\text{in},0}$, and $\mathbb{E}[\text{LOS}_i]$ across all 100 replications (Eq. 13); and (Panel B) a 20-cell parametric heatmap over severity–LOS coupling $\beta \in \{0.0, 0.6, 1.2, 1.8, 2.4\}$ and queue deterioration rate $\alpha_{\text{det}} \in \{0.05, 0.18, 0.35, 0.50\}$ day^{-1}.

5.4.1. Robustness to Multi-Parameter Clinical Noise

As shown in Figure 4A, when prognostic probabilities and expected stay estimates are perturbed by Gaussian noise, all five clinical policies degrade gracefully while remaining strictly superior to chronological FCFS (431.14 ± 1.13 survivors, which requires no clinical information). At moderate estimation noise ($\sigma_{\text{noise}} = 0.10$), MGS (446.92 ± 1.22, $\Delta S = +15.79, p_{\text{Holm}} < 0.001$), CMU-MGS (446.82 ± 1.20, $\Delta S = +15.68, p_{\text{Holm}} < 0.001$), and AUSTERE (446.28 ± 1.21, $\Delta S = +15.15, p_{\text{Holm}} < 0.001$) retain over 75% of their noise-free survival advantage over FCFS. Under severe noise ($\sigma_{\text{noise}} = 0.20$), because CMU-MGS compounds ratio estimation errors in both numerator ($\widetilde{\Delta P}_i$) and denominator ($\widetilde{\mathbb{E}}[\text{LOS}_i]$), single-ratio MGS edges ahead slightly (444.00 vs. 443.40 survivors), though both still save over +12.2 expected lives relative to FCFS ($p_{\text{Holm}} < 0.001$). Consistent with Argon and Ziya [20], simpler index structures become attractive when both prognostic and duration estimates are highly imprecise.

5.4.2. When Does $c\mu$ Triage Outperform Greedy Marginal Gain?

Figure 4B resolves the structural relationship between greedy MGS and duration-normalized CMU-MGS across clinical regimes:

- Severity-Independent LOS ($\beta = 0.0$): When all patients in pool j share the same expected stay $\mathbb{E}[\text{LOS}_i] = \mu_{\text{los},j}$ regardless of severity, the denominator in Eq. (11) is constant within each pool. Consequently, CMU-MGS and MGS induce identical priority orderings ($\Delta S_{\text{CMU–MGS}} = 0.00$ across all α_{det}).

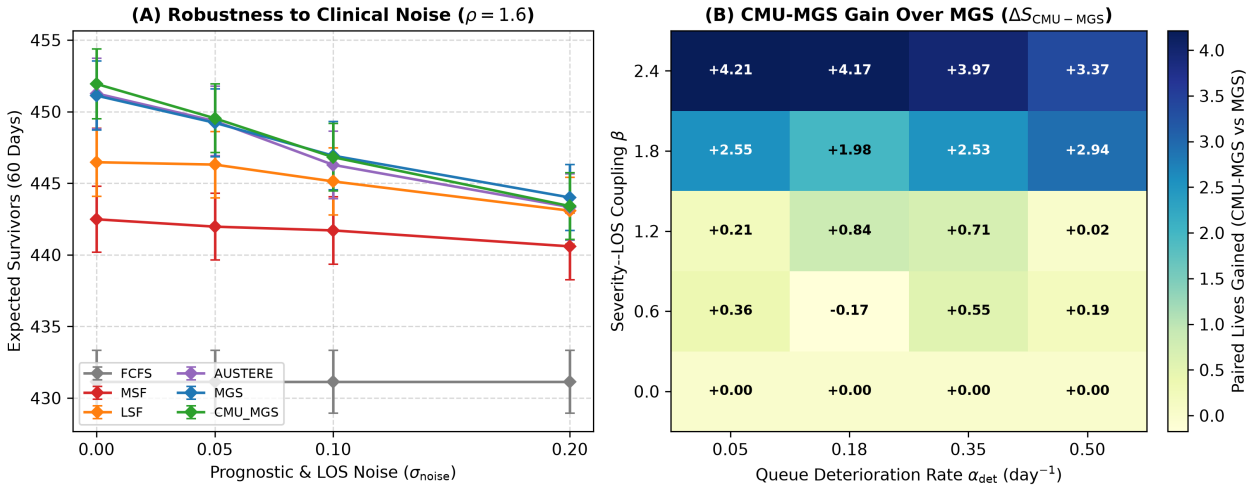


Figure 4: Sensitivity analysis at severe congestion ($\rho = 1.60$): (A) expected survivors across 100 replications under simultaneous observation noise $\sigma_{\text{noise}} \in \{0.00, 0.05, 0.10, 0.20\}$ affecting P_{out}, $P_{\text{in},0}$, and $\mathbb{E}[\text{LOS}]$; (B) parametric heatmap of the paired survival gain of CMU-MGS over greedy MGS ($\Delta S_{\text{CMU–MGS}}$) across severity–LOS coupling β and queue deterioration rate α_{det}.

- Weak Severity–LOS Coupling ($\beta = 0.6$): When LOS varies modestly with severity, CMU-MGS and MGS perform similarly ($\Delta S_{\text{CMU–MGS}} \in [-0.17, +0.55]$ lives), indicating that greedy marginal survival gain is already near-optimal.

- Strong Severity–LOS Coupling ($\beta \in \{1.8, 2.4\}$): When critically ill patients with low P^i_{out} require substantially longer stays than moderately ill candidates, greedy MGS repeatedly locks beds with long-stay admissions. In this regime, CMU-MGS achieves highly significant gains over MGS across all deterioration rates, saving +1.98 to +2.94 additional lives at $\beta = 1.8$ ($t \in [5.47, 7.77], p < 0.001$) and +3.37 to +4.21 additional lives at $\beta = 2.4$ ($t \in [8.32, 10.20], p < 0.001$).

- Impact of Rapid Queue Deterioration ($\alpha_{\text{det}} \in \{0.35, 0.50\}$): As pre-ICU deterioration accelerates, the total survival advantage of CMU-MGS over FCFS expands dramatically—from +15.32 lives at ($\beta = 1.2, \alpha_{\text{det}} = 0.05$) to +31.83 lives at ($\beta = 1.2, \alpha_{\text{det}} = 0.50$)—because chronological queueing allows high-benefit candidates to decay rapidly while waiting behind earlier arrivals.

6. Limitations and Ethical Considerations

Translating operations research models into clinical triage guidelines requires explicit recognition of modeling boundaries and bioethical constraints:

(1) Empirical Calibration vs. Patient-Level Electronic Health Records: While our bed capacities are taken directly from DATASUS/CNES for HU-UFMA and specialty-level LOS and mortality distributions are calibrated from published national registries (AMIB/Epimed and RBPN), individual patient trajectories are simulated parametrically rather than replayed from de-identified patient-level electronic health records (EHR). Future clinical deployment requires fitting joint survival–duration copulas directly to local hospital EHR streams.

(2) Prognostic Uncertainty and Algorithmic Equity: Efficiency-oriented policies (MGS, CMU-MGS, AUSTERE) rely on prognostic survival and LOS estimates. In clinical practice, severity scores such as SOFA and SAPS 3 exhibit calibration drift across demographic and socioeconomic groups [5, 2]. Furthermore, pure $c\mu$ prioritization systematically penalizes patients with chronic comorbidities or severe multi-organ dysfunction whose expected recovery time $\mathbb{E}[\text{LOS}_i]$ is protracted. Triage committees should therefore embed index scores within multi-principle frameworks [5] that incorporate equity safeguards, age/life-cycle considerations, and periodic clinical reassessment rather than delegating admission decisions to an automated score alone.

(3) Operational Scope: Our discrete-event engine assumes stationary Poisson arrival rates within each 60-day regime, instantaneous bed turnaround upon discharge (omitting 2–4 hour sanitization delays), and no premature step-down discharge (demand-driven early discharge as in Chan et al. [9]). Incorporating non-homogeneous epidemic wave arrivals and dynamic step-down transfers represents a natural extension.

7. Conclusions

This paper developed and empirically calibrated a continuous-time discrete-event simulation and retro-

spective MILP benchmarking framework for dynamic multi-type ICU bed allocation under resource scarcity. Grounded in the 55-bed critical care structure of HU-UFMA (DATASUS/CNES) and specialty-specific parameters from Brazilian intensive care registries, our study yields four main conclusions:

(1) Value of Dynamic Triage Under Scarcity: While admission rules have negligible practical impact when capacity is ample ($\rho = 0.70$, relative effect $< 0.1\%$), priority indices that account for marginal survival benefit and bed-time consumption (CMU-MGS and AUSTERE) save 20.13–20.80 more expected lives than FCFS over a 60-day surge ($\rho = 1.60$, +4.82%, $p_{\text{Holm}} < 0.001$) and reduce admitted patient waiting times by nearly half (9.56 vs. 18.27 hours).

(2) Regime-Dependent Advantage of $c\mu$ Normalization: Our parametric sensitivity analysis demonstrates that greedy marginal survival triage (MGS) suffices when length of stay is weakly correlated with severity ($\beta \leq 0.6$). However, when severe patients require substantially longer ICU stays ($\beta \geq 1.8$), normalizing marginal benefit by expected bed-days (CMU-MGS) saves up to +4.21 additional expected lives over MGS ($p < 0.001$) by accelerating bed turnover.

(3) Targeted Benefit of Partial Bed Pooling: Hierarchical overflow sharing between adult general and coronary care units provides substantial survival gains at moderate-to-balanced congestion ($\rho = 1.00$, saving +3.46 to +3.90 expected lives, $p < 0.001$), whereas its benefit vanishes under extreme overload ($\rho = 1.60$) once both units saturate simultaneously.

(4) Informational Robustness and Offline Benchmark: Even under simultaneous Gaussian estimation noise ($\sigma_{\text{noise}} = 0.20$) across all survival and LOS inputs, clinical index rules preserve a highly significant advantage of +12.2 to +12.9 expected lives over FCFS ($p_{\text{Holm}} < 0.001$).

Data and Code Availability

All simulation source code, unit tests, DATASUS/CNES extraction files, configuration scripts, raw replication outputs (runs.csv, offline_ub.csv, and sensitivity grids), and plotting scripts are openly available in the project's public repository at https://github.com/MODAL-UFMA/alocacao-uti-web.

CRediT Authorship Contribution Statement

Carlos Eduardo Veras Gomes: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft. João Augusto Ferreira Nunes de Carvalho: Methodology, Software, Data curation, Visualization, Writing – original draft. Francisco Glaubos Nunes Clímaco: Conceptualization, Methodology, Formal analysis, Supervision, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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